Generalized Burau representations and quantum gates: arrangements, compact closures, and prior results

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Abstract

We collect explicit generalized Burau matrices, their invariant Hermitian forms, and a quantum-gate interpretation for finite Coxeter arrangements. We determine the conditions under which the invariant Hermitian forms are positive definite in terms of the Coxeter exponents. A direct rank-two proof gives a compact-closure formula for $0 < |\theta| < \min\{\pi/h, \pi/10\}$, including roots of unity. For E_6, E_7, E_8, this formula holds throughout the parameter interval where the invariant Hermitian form is positive definite, except at the undeformed Coxeter specialization. We explicitly show that these exceptional-type closure conclusions also follow from the existing two-eigenvalue classification of Freedman–Larsen–Wang; they are not claimed as new density or finite-image classification theorems. In dimension eight, the resulting gates are non-Clifford, and are projectively universal away from the Coxeter specialization. Appendices retain the Dynkin-type dictionary, algebraic convolution formulas for general affine arrangements, and a six-line based-monodromy example from the earlier note. The contribution of this document is an explicit synthesis with proofs, conventions, and reproducible calculations; theorem-by-theorem overlap with prior work is recorded.

1 Introduction and statement

In earlier work, we proposed a method for constructing quantum gates from monodromy representations obtained by applying middle convolution to KZ-type systems [10, 11]. Motivated by this construction, the present note studies generalized Burau representations of Artin–Tits groups and their use in quantum gate construction. Starting from Squier's representations [1], we determine explicit conditions for positive definiteness of the invariant Hermitian forms in finite Coxeter type and describe the resulting unitary gates. We also recall the middle-convolution construction for general affine hyperplane arrangements on their directional closures [20, 21] and make its relation to Burau pullbacks explicit in an equal-weight specialization (Appendix D).

Our emphasis is on explicit parameter ranges and on the distinction between finite-order generators and a finite generated group.

General Coxeter type versus arbitrary arrangements. For any finite-rank Coxeter matrix, not necessarily of finite type, the known construction on $\mathbb{C}(t)^r$ is

$$R_i e_j = \begin{cases} -t^2 e_i, & j = i, \\ e_j + 2t\cos(\pi/m_{ij}) e_i, & j \neq i, \end{cases} \qquad \cos(\pi/\infty) = 1. \tag{1}$$

The matrices satisfy the Artin relations, have determinant $-t^2$, and satisfy $(R_i - I)(R_i + t^2 I) = 0$; they specialize for every $t \neq 0$ [1, 2]. Our parameter is the negative of that in [2, Definition 2.1], and agrees with [17]. An arbitrary affine arrangement need not have a Coxeter

matrix or a Dynkin type. Appendix A explains the type notation; Appendices B–G treat a different, direction-dependent construction for arbitrary arrangements. No identification of the exceptional Coxeter representations with a middle-convolution output is assumed.

The closure results below require finite irreducible Coxeter type.

Let W be a finite irreducible Coxeter group of rank $r \geq 2$, with Coxeter matrix (m_{ij}), Coxeter number h, and Artin group B_W. Write $P_W = \ker(B_W \to W)$. Define the real symmetric matrix

$$A_{ii} = 0, \qquad A_{ij} = 2\cos(\pi/m_{ij}) \quad (i \neq j). \tag{2}$$

For $t = e^{i\theta}$ put

$$H_\theta = 2\cos\theta\, I - A, \qquad R_i = I - te_ie_i^* H_\theta. \tag{3}$$

These are the generalized Burau matrices, with $R_ie_i = -t^2e_i$ and $R_ie_j = e_j + tA_{ij}e_i$ for $j \neq i$. When $H_\theta > 0$, let

$$S = H_\theta^{1/2}, \qquad U_i = SR_iS^{-1}, \qquad \lambda = -e^{2i\theta}, \qquad D_\theta = \overline{\langle\lambda\rangle} \subset \mathrm{U}(1). \tag{4}$$

Closures throughout the paper are in the ordinary compact-group topology.

Theorem 1.1 (Explicit closure formula)**.** *Suppose*

$$0 < |\theta| < \min\{\pi/h, \pi/10\}. \tag{5}$$

Then the unitarized representation satisfies

$$\overline{\rho_\theta(B_W)} = \{g \in \mathrm{U}(r) : \det g \in D_\theta\}, \tag{6}$$

$$\overline{\rho_\theta(P_W)} = \{g \in \mathrm{U}(r) : \det g \in \overline{\langle\lambda^2\rangle}\}. \tag{7}$$

In particular, both projective images are dense in $\mathrm{PU}(r)$*. If* θ/π *is irrational, both closures are* $\mathrm{U}(r)$*. If* λ *has finite order* k*, the closures have respectively* k *and* $k/\gcd(k,2)$ *connected components, each a coset of* $\mathrm{SU}(r)$*.*

For $h \geq 10$*, the hypothesis covers the full positive interval* $0 < |\theta| < \pi/h$*. In* $|\theta| < \pi/h$ *the Artin image is then finite if and only if* $\theta = 0$*; at* 0 *the image is the reflection representation of* W*, and the pure image is trivial.*

The exceptional types have $h = 12, 18, 30$, respectively. Thus the theorem is exhaustive within their positive intervals. It is not a classification of specializations outside those intervals, or at degenerate endpoints.

Relation to prior work. The representation and form are due to Squier [1]. The exceptional-type conclusion of Theorem 1.1 is a consequence of Freedman–Larsen–Wang [13]; the precise reduction is given in Proposition 6.1. Thus an explicit interval or a root-of-unity specialization does not, by itself, establish novelty. Marin's generic compact specialization theorem [4, Theorem 4] and the rank-two analysis of Funar–Kohno [5] provide further precedent. The finite $W(E_8)$ connection with three-qubit gates is also known [7, 8]. Selective-phase gate descriptions and the Toffoli comparison overlap with the author's separate working draft [9, Section 6]. Section 6 distinguishes known constructions, consequences of existing theorems, and explicit calculations.

2 Invariant forms and explicit unitary gates

The matrix in (3) is the specialization of Squier's two-variable form obtained by setting both variables equal to t and dividing the Gram matrix by t. The matrices R_i satisfy the Artin relations [1]. In our convention, type A_r corresponds to reduced Burau with classical parameter $q = t^2$.

Proposition 2.1. *For $|t| = 1$, $R_i^* H_\theta R_i = H_\theta$. If $d_1, \dots, d_r$ are the Coxeter exponents, the eigenvalues of H_θ are*

$$2\cos\theta - 2\cos(\pi d_j/h), \qquad 1 \le j \le r. \tag{8}$$

For $-\pi < \theta \le \pi$, this particular form is positive definite exactly when $|\theta| < \pi/h$.

Proof. Set $E_i = e_i e_i^*$. Expansion gives

$$R_i^* H_\theta R_i - H_\theta = [-t - \bar{t} + |t|^2 (H_\theta)_{ii}] H_\theta E_i H_\theta = 0.$$

The spectrum of A is $2\cos(\pi d_j/h)$; for crystallographic types this is the standard Cartan/adjacency spectral identity, with diagonal symmetrization in the nonsimply-laced cases [3]. The finite Coxeter version uses the same geometric Gram matrix. The smallest exponent is 1, so the smallest eigenvalue of H_θ is $2\cos\theta - 2\cos(\pi/h)$. □

At a degenerate parameter, $\ker H_\theta$ is fixed pointwise by all R_i and the form descends to a nondegenerate form on the quotient. We do not use that quotient in Theorem 1.1. The negative-definite region can also be unitarized after changing the overall sign; our finite-image statement is explicitly restricted to the positive interval.

In the positive interval set

$$p_i = \frac{S e_i}{\sqrt{2\cos\theta}}, \qquad U_i = I + (\lambda - 1) p_i p_i^*, \qquad \langle p_i, p_j \rangle = -\frac{A_{ij}}{2\cos\theta} \quad (i \ne j). \tag{9}$$

The p_i form a basis of unit vectors. Each U_i is the identity on $p_i^\perp$ and multiplies p_i by λ; hence

$$\det U_i = \lambda, \qquad \operatorname{Tr} U_i = r - 1 + \lambda. \tag{10}$$

The same positive square root S is used for every generator.

For E_r, $r = 6, 7, 8$, use the edges

$$\mathcal{E}_r = \{\{i, i+1\} : 1 \le i \le r-2\} \cup \{\{3, r\}\}.$$

Then A is the ordinary adjacency matrix. This convention fixes all matrices without a choice of root coordinates.

Type	h	Exponents	Positive interval
E_6	12	$1, 4, 5, 7, 8, 11$	$\lvert\theta\rvert < \pi/12$
E_7	18	$1, 5, 7, 9, 11, 13, 17$	$\lvert\theta\rvert < \pi/18$
E_8	30	$1, 7, 11, 13, 17, 19, 23, 29$	$\lvert\theta\rvert < \pi/30$

For example, $U_3 = I + (\lambda - 1)(Se_3)(Se_3)^*/(2\cos\theta)$. In the original basis, row 3 of R_3 has entries t in columns $2, 4, r$, entry $-t^2$ in column 3, and zeros elsewhere; all other rows are those of the identity.

3 From two-dimensional subspaces to the full closure

The proof below does not infer density from irreducibility or from infinite order alone.

Lemma 3.1. *Let p, q be independent unit vectors with $0 < |\langle p, q \rangle| < 1$. On $T = \operatorname{span}\{p, q\}$ consider $V_p = I + (\lambda - 1) pp^*$ and $V_q = I + (\lambda - 1) qq^*$, where $\lambda = -e^{2i\theta}$ and $0 < |\theta| < \pi/10$. Their projective image is dense in $\mathrm{PU}(T)$, and their unitary closure contains $\mathrm{SU}(T)$.*

Proof. Under $\mathrm{PU}(2) \simeq \mathrm{SO}(3)$, each generator is a rotation of angle $\alpha = \pi - 2|\theta| \in (4\pi/5, \pi)$. Their axes are distinct: equality of the unoriented axes would mean that the two eigenlines are equal or orthogonal, both excluded by hypothesis.

The finite rotation groups are cyclic, dihedral, tetrahedral, octahedral, and icosahedral. In the first two, every rotation with angle different from π has the same axis. In the last three, every angle smaller than π is at most $4\pi/5$. Thus the projective group is infinite. An infinite proper closed subgroup of $\mathrm{SO}(3)$ has identity component a circle and is contained in its normalizer. In that normalizer all rotations other than half turns have the circle's axis, again a contradiction. The projective closure is therefore $\mathrm{SO}(3)$.

If $\mathfrak{k} \subset \mathfrak{u}(2)$ is the Lie algebra of the unitary closure, its projection onto $\mathfrak{pu}(2)$ is surjective. Hence $[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{su}(2)$ projects onto $\mathfrak{su}(2)$ and equals it. The closure contains $\mathrm{SU}(2)$. □

Lemma 3.2. *Let $v_1, \dots, v_r$ be a basis of a Hilbert space V, and let Γ be a connected graph on these indices. A closed subgroup of $\mathrm{U}(V)$ containing $\mathrm{SU}(\mathrm{span}\{v_i, v_j\})$, acting trivially on its orthogonal complement, for every edge $\{i, j\}$ contains $\mathrm{SU}(V)$.*

Proof. We use a connected ordering of a spanning tree. Suppose $\mathrm{SU}(S)$ has already been obtained and an edge supplies $\mathrm{SU}(T)$, where $\dim T = 2$ and $S \cap T = \mathbb{C}u$ for a unit vector u. Choose a unit vector $w \in T \cap u^\perp$ and write $w = a + bz$, with $a \in S \cap u^\perp$, $z \perp S$ a unit vector, and $b \neq 0$. In the complexified generated Lie algebra, uw^* belongs to $\mathfrak{sl}(T)$ and ua^* to $\mathfrak{sl}(S)$. Subtraction gives uz^*; similarly one obtains zu^*. Commutators with $\mathfrak{sl}(S)$ give xz^*, zx^* for all $x \in S$, and their commutators give the missing trace-zero diagonal operator. They generate $\mathfrak{sl}(S + T)$. The original compact Lie algebra therefore contains $\mathfrak{su}(S + T)$. Induction, starting from one edge, gives the assertion. □

Proof of Theorem 1.1. For an edge of the Coxeter graph, (9) and positivity give $0 < |\langle p_i, p_j \rangle| < 1$. The two gates preserve their span and fix its orthogonal complement pointwise. Lemma 3.1 gives an embedded $\mathrm{SU}(2)$ in the closure on each edge. Connectedness and Lemma 3.2 imply that the full closure G contains $\mathrm{SU}(r)$.

The determinant of every word is a power of λ. By compactness, $\det G = \overline{\langle \lambda \rangle} = D_\theta$. Since the kernel of determinant is contained in G, this proves (6).

The pure group has finite index in the Artin group. Its image closure G_P consequently has finite index in G, so contains G° and in particular $\mathrm{SU}(r)$. The exponent-sum homomorphism of the Artin group is even on P_W, since its parity is the Coxeter sign character. Moreover $\sigma_i^2 \in P_W$ has exponent sum 2. Hence the determinant image of P_W is exactly $\langle \lambda^2 \rangle$, proving (7). The assertions about rational and irrational parameters follow from closed subgroups of $\mathrm{U}(1)$.

At $\theta = 0$, the matrices R_i are the faithful geometric reflection representation of W. Every other parameter in (5) has closure containing the positive-dimensional group $\mathrm{SU}(r)$ and therefore has infinite image. If $h \geq 10$, the positive interval is contained in $|\theta| < \pi/10$. □

Remark 3.3 (Scope for other types). For crystallographic groups, the full-positive-interval conclusion also applies to A_n with $n \geq 9$, $B_n = C_n$ with $n \geq 5$, D_n with $n \geq 6$, and F_4. Smaller Coxeter numbers are still covered on the smaller interval in (5); no assertion is made here about their remaining positive parameters. Merely assuming that λ is a root of unity would not prove finite image.

4 Clifford obstruction and quantum universality

We fix the usual identification $\mathbb{C}^8 = (\mathbb{C}^2)^{\otimes 3}$ in the orthonormal coordinates of (4). The Clifford group means the normalizer of the tensor-product Pauli group, with arbitrary global phases allowed. Its binary symplectic action is standard [6].

Lemma 4.1. *If C is a nonscalar Clifford unitary on n qubits and $d = 2^n$, then*

$$|\operatorname{Tr} C|^2 \le d^2/2. \tag{11}$$

Consequently, for $n \ge 3$ and $\lambda \ne 1$, no unitary with spectrum $\{1^{(d-1)}, \lambda\}$ is Clifford, in any orthonormal encoding.

Proof. Let $F \in \mathrm{Sp}(2n, 2)$ be the induced permutation of Pauli labels. The conjugation operator Ad_C is a signed permutation on the Hermitian Pauli basis, and $\operatorname{Tr}(\mathrm{Ad}_C) = |\operatorname{Tr} C|^2$. If $F \ne I$, at most 2^{2n-1} labels are fixed, so its trace is at most that number. If $F = I$, the Clifford is a Pauli up to phase; unless scalar, its trace is zero. This proves (11). For the stated spectrum, $|\operatorname{Tr} U| \ge d - 2$, whereas $(d-2)^2 > d^2/2$ for $d \ge 8$. Trace magnitude is unchanged by a global phase or a common unitary change of encoding. □

Corollary 4.2. *For type E_8 and $|\theta| < \pi/30$, every generator U_i is non-Clifford. For $0 < |\theta| < \pi/30$, the gates $\{U_i, U_i^{-1} : 1 \le i \le 8\}$ are approximately universal on three qubits, up to global phase. At $\theta = 0$ all generators remain non-Clifford, but the generated group is the finite group $W(E_8)$.*

Proof. Here $\lambda \ne 1$ and $\operatorname{Tr} U_i = 7 + \lambda$, so $|\operatorname{Tr} U_i|^2 \ge 36 > 32$. Apply Lemma 4.1 and the projective closure in Theorem 1.1. □

Non-Cliffordness alone is thus insufficient for universality, even within this single continuous family. A fixed three-qubit gate set is meant here. If its gates can be applied to arbitrary selected triples in a larger register, one obtains circuit universality by approximating arbitrary one- and two-qubit gates inside a triple [12]. No implementation cost or fault-tolerance advantage is asserted.

An exact root-of-unity example. Take $\theta = \pi/60$. Then

$$t = e^{i\pi/60}, \qquad \lambda = e^{2\pi i\, 31/60}, \qquad U_i^{60} = I.$$

Although all eight gates have exact order 60, their group is infinite, its projective closure is $\mathrm{PU}(8)$, and its full closure is

$$\{g \in \mathrm{U}(8) : \det g \in \mu_{60}\}.$$

The pure-group closure has determinant in μ_{30}. Inverses are already positive powers of the gates in this example.

Gate realization. If a unitary V_i satisfies $V_i|111\rangle = p_i$, then

$$U_i = V_i \operatorname{diag}(1, 1, 1, 1, 1, 1, 1, \lambda) V_i^*. \tag{12}$$

Thus each generator is a conjugated three-qubit controlled phase. Such a V_i can be constructed by completing p_i to an orthonormal basis, or by a Householder map. Formula (12) is a matrix realization, not a claim that V_i is a free or Clifford operation. At $\theta = 0$ it is a conjugate of CCZ, hence also of Toffoli. This is an individual conjugacy statement, not a simultaneous identification of the entire generator set with one gate.

The other exceptional dimensions. Types E_6, E_7 give projectively universal gates on $\mathbb{C}^6, \mathbb{C}^7$. One may use $\mathbb{C}^6 \simeq \mathbb{C}^2 \otimes \mathbb{C}^3$, or encode either space in three qubits. Under the latter choice, the extension $U_i \oplus I_{8-r}$ is still a nontrivial rank-one phase gate and is non-Clifford by Lemma 4.1. The group generated by these extensions preserves the code and the unused subspace, so is not universal on all of $\mathbb{C}^8$.

5 Arrangement interpretation and reproducibility

Let X_W be the complement of the complexified reflection arrangement. Then $P_W = \pi_1(X_W)$ and $B_W = \pi_1(X_W/W)$. Formula (7) therefore gives the compact closure for the actual arrangement-complement representation, not only for the quotient by W. For a general arrangement and a map f into X_W, a pullback representation inherits the form, but its image can be a much smaller subgroup. Density does not follow without information about $f_*(\pi_1)$.

An arbitrary arrangement need not have a Dynkin type. The results above concern reflection arrangements and do not supply a canonical universal gate construction for all arrangements. We also do not identify these representations with a Katz–Long–Moody quotient; such a comparison is unnecessary for the closure proof.

The accompanying `verify_coxeter_gates.py` constructs the matrices and exports a concrete E_8 gate set at $\theta = \pi/60$. It verifies the Artin and invariant-form identities exactly over $\mathbb{Z}[t, t^{-1}]$ for E_6, E_7, E_8, and checks the positive square-root construction, spectral formulas, determinants, and Clifford trace obstruction numerically. The numerical checks are consistency checks; the density, finite-image classification, and non-Clifford conclusions follow from the proofs above rather than finite word searches.

6 Comparison with existing theorems

The following reduction is included to make the overlap a mathematical statement, rather than an inference from similar titles or abstracts.

Proposition 6.1 (Exceptional-type closure as an existing-theorem consequence)**.** *For $W = E_r$, $r = 6, 7, 8$, and $0 < |\theta| < \pi/h$, the inclusion $\mathrm{SU}(r) \subset \overline{\rho_\theta(B_W)}$ follows from [13, Theorems 1.1, 1.4 and 1.6], without using the rank-two density proof of Section 3.*

Proof. Put $G = \overline{\langle U_1, \dots, U_r \rangle}$. This is a compact matrix group acting faithfully. It acts irreducibly: a nonzero invariant subspace contains some p_i, since the p_i span and $(U_i - I)v = (\lambda - 1)p_i\langle p_i, v\rangle$. Nonzero overlaps along the connected Coxeter diagram then put every p_j in that subspace.

All Artin generators in a connected simply-laced diagram are conjugate: if $aba = bab$, then $(ab)a(ab)^{-1} = b$. Consequently the conjugacy class of U_1 generates G topologically. Its eigenvalues are 1 and λ with multiplicities $r - 1$ and 1, and $\lambda \neq \pm 1$. These are exactly the two-eigenvalue hypotheses of [13].

If $G/Z(G)$ were finite, [13, Theorem 1.6] would imply $U_1^m \in Z(G)$ for some $m \in \{3, 4, 5\}$. By irreducibility this forces $\lambda^m = 1$. But the smaller angle of λ is $\pi - 2|\theta| \in (4\pi/5, \pi)$, which excludes those orders. Thus $G/Z(G)$ has positive dimension.

Let $K = [G^\circ, G^\circ]$. Applying [13, Theorems 1.1 and 1.4], K acts irreducibly, $G = G^\circ Z(G)$, and its representation belongs to their list. In dimensions $6, 7, 8$, the alternatives to the defining or dual representation of $\mathrm{SU}(r)$ all preserve a nondegenerate complex bilinear form. Explicitly, dimension 6 also permits $\bigwedge^2 \mathbb{C}^4$ for $\mathrm{SU}(4)$ and the defining $\mathrm{Sp}(6)$ module; dimension 7 has no alternative; dimension 8 also permits the defining $\mathrm{Sp}(8)$ module, the spin module for $\mathrm{Spin}(7)$, and the vector and half-spin modules for $\mathrm{Spin}(8)$. Here $\mathrm{Sp}(d)$ denotes the compact symplectic group in complex dimension d.

If such a bilinear form B existed, the center of G° would act by scalars, and $G = G^\circ Z(G)$ would imply $U_i^{\mathsf{T}} B U_i = cB$ for some $c \neq 0$. The eigenvalue multiset would then be invariant under $z \mapsto c/z$. The multiplicities $r - 1$ and 1 differ, so 1 must map to itself and λ to itself. Thus $c = 1$ and $\lambda^2 = 1$, a contradiction. Only the defining $\mathrm{SU}(r)$ image remains, as required. □

The determinant and finite-index arguments in the proof of Theorem 1.1 now give the full and pure closure formulas. The exceptional-type finite-image statement and projective universality are therefore further consequences, not independent classification results. This reduction does not assert that the same formulas occur verbatim in [13]. For nonsimply-laced types, one cannot use the single-conjugacy-class step without further justification; the direct proof in Section 3 still applies in its stated interval.

Claim-by-claim status

Item in this note	Prior result or deduction	Status used here
Generalized Burau matrices and invariant form	Squier [1]; modern convention in [2].	Known construction, recalled explicitly.
Dynkin equations, ranks, exponents, and positivity	Root-system data and the Cartan spectral identity [14, 3].	Standard data and a direct spectral consequence.
E_6, E_7, E_8 compact closure	Proposition 6.1, applying [13].	Existing-theorem consequence; the rank-two proof is an alternative derivation.
Full unitary components and pure-group closure	Determinant image and finite index, once $\mathrm{SU}(r)$ is contained.	Elementary consequences of the closure statement.
Finite images in the positive E interval	Infinite projective closure for $\theta \neq 0$; Coxeter image at 0.	A consequence within the stated interval, not a general classification.
Non-Clifford generators	Pauli/symplectic formalism [6] and the trace estimate in Lemma 4.1.	Elementary consequence. An identical published formulation has not been located; no priority claim is made.
Three-qubit universality and finite E_8 limit	Closure implies universality; finite $W(E_8)$ gate realizations precede this note [7, 8].	Application and explicit family; not a new universality criterion.
General-arrangement convolution and middle quotient	Existing constructions [20, 21, 22].	Recalled formulas with hypotheses; not a new functor.
Six-line monodromy and descent calculation	Burau pullback, fixed based loops, and a routine module quotient.	Explicit computation. Independent novelty of the particular presentation is unestablished.

Parameter and scope checks. Marin's compact specialization result assumes a transcendental parameter on the unit circle sufficiently near 1 [4, Theorem 4]; it cannot alone be quoted for every algebraic parameter in our interval. That limitation does not rescue novelty of the exceptional-type result, because Proposition 6.1 supplies a different existing theorem. Funar–Kohno analyze rank-two root-of-unity images [5]; parameter sign conventions must be matched before transferring their exceptional orders. A general arrangement pullback need not have the same image as the whole braid group, as already noted above.

Research status. This is an expository and computational note. The explicit gate matrices, the common unitarization, and the direct closure proof are useful to retain, but do not by themselves demonstrate a new principal theorem. The literature comparison is targeted to the

claims made here, rather than a claim of exhaustive priority clearance. No new theorem is inferred merely from failure to locate an identical wording, parameterization, or example.

A Writing reflection arrangements by Dynkin type

The scope of the type notation

A general affine arrangement is specified by equations

$$\mathcal{A} = \{H_a : \ell_a(x) = b_a\}_{a=1}^{N}, \qquad Q_{\mathcal{A}}(x) = \prod_{a=1}^{N} (\ell_a(x) - b_a).$$

There is no rule assigning a Dynkin type to every such arrangement. The notation "type X_r" used here means that, after a linear change of coordinates and removal of any redundant ambient factor, the arrangement is the complexified reflection arrangement of a root system $\Phi(X_r)$. A common translation is allowed if the original equations are affine. In particular, counting hyperplanes or drawing their intersection graph does not determine a Dynkin type.

For a reduced crystallographic root system $\Phi \subset V_{\mathbb{R}}^*$, the construction is

$$\boxed{\mathcal{A}(X_r) = \{\ker \alpha : \alpha \in \Phi^+(X_r)\}, \qquad Q_{X_r}(x) = \prod_{\alpha \in \Phi^+(X_r)} \alpha(x).} \tag{13}$$

Here $V = V_{\mathbb{R}} \otimes \mathbb{C}$, and one positive root is used from each pair $\{\alpha, -\alpha\}$. Thus the Dynkin diagram has r vertices, whereas the arrangement has $|\Phi|/2$ hyperplanes. The root-system construction is implemented in [15].

The irreducible crystallographic types are $A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, G_2$; reducible types give products of arrangements. See [14, Theorem 23.7] for the classification. Finite real reflection arrangements also include the noncrystallographic Coxeter types $H_3, H_4, I_2(m)$, for which one uses Coxeter rather than crystallographic Dynkin data.

Classical types in coordinates

For A_n, use $V = \{(x_1, \ldots, x_{n+1}) \in \mathbb{C}^{n+1} : \sum_i x_i = 0\}$. For B_n, C_n, D_n, use $V = \mathbb{C}^n$. In the table, every displayed linear form is set equal to zero, with $i < j$ in the appropriate coordinate range.

Type	Defining linear forms	Number of hyperplanes
A_n	$x_i - x_j$	$n(n+1)/2$
B_n	$x_i,\ x_i - x_j,\ x_i + x_j$	n^2
C_n	$2x_i,\ x_i - x_j,\ x_i + x_j$	n^2
D_n	$x_i - x_j,\ x_i + x_j$	$n(n-1)$

These follow by taking kernels of the standard classical roots [14, §20.3]; the B_n equations are also listed explicitly in [16]. Consequently, up to nonzero constants,

$$\begin{aligned} Q_{A_n} &= \prod_{1 \le i < j \le n+1} (x_i - x_j), \\ Q_{B_n} = Q_{C_n} &= \left(\prod_{i=1}^{n} x_i \right) \prod_{1 \le i < j \le n} (x_i^2 - x_j^2), \\ Q_{D_n} &= \prod_{1 \le i < j \le n} (x_i^2 - x_j^2). \end{aligned}$$

The A_n polynomial is restricted to V above; using all of $\mathbb{C}^{n+1}$ instead gives the familiar nonessential braid arrangement with a redundant diagonal direction.

For example,

$$\mathcal{A}(B_3) = \{x_1 = 0, x_2 = 0, x_3 = 0\} \cup \{x_i - x_j = 0, x_i + x_j = 0 : 1 \le i < j \le 3\}$$

has nine planes, even though its Dynkin diagram has three vertices. The arrangement of type D_3 has six planes and is isomorphic to type A_3; the irreducible D_n family is usually indexed by $n \ge 4$.

Why B_n and C_n give the same arrangement. Their root systems use $\pm e_i$ and $\pm 2e_i$, respectively, together with $\pm e_i \pm e_j$. Multiplying a normal by 2 does not change its kernel. Hence

$$\mathcal{A}(B_n) = \mathcal{A}(C_n)$$

in these coordinates. The hyperplane set alone therefore does not retain the distinction between these dual root systems or the direction of the double-edge arrow in their Dynkin diagrams.

Recovering the E-type equations from the diagram

One need not list coordinates in $\mathbb{R}^8$ to specify the exceptional arrangements. Choose simple roots $\alpha_1, \dots, \alpha_r$, and use the coordinates $z_i = \alpha_i(x)$. If a positive root has simple-root coefficients $c = (c_1, \dots, c_r)$, its hyperplane is exactly

$$H_c = \{c_1 z_1 + \cdots + c_r z_r = 0\}. \tag{14}$$

For E_r, $r = 6, 7, 8$, use the diagram with edge set

$$\mathcal{E}_r = \{\{i, i+1\} : 1 \le i \le r-2\} \cup \{\{3, r\}\}.$$

For $r = 6$ this agrees with the numbering used in Section 1. Its Cartan matrix is

$$C_{ii} = 2, \qquad C_{ij} = -1 \text{ if } \{i, j\} \in \mathcal{E}_r, \qquad C_{ij} = 0 \text{ otherwise.}$$

Start with the unit vectors e_i in the simple-root coefficient lattice and repeatedly apply

$$s_i(c) = c - \left(\sum_j C_{ij} c_j\right) e_i. \tag{15}$$

Closing under these reflections gives all roots; retain those whose coefficients are all nonnegative, and apply (14). This is the usual simple-reflection generation of a finite root system [14, §§21–23], written as an explicit equation-generation procedure.

The following counts distinguish arrangement size from rank:

Type	Rank	Roots	Hyperplanes
E_6	6	72	36
E_7	7	126	63
E_8	8	240	120
F_4	4	48	24
G_2	2	12	6

The three E-type counts were also checked in the original note by the finite orbit procedure (15). The same root-kernel recipe (13) applies to F_4 and G_2.

From the Dynkin diagram to the Burau matrices

For the representation in Section 1, the relevant data are the Coxeter labels. The translation is

Dynkin adjacency	m_{ij}	Coefficient of e_i in $R_i e_j$
No edge	2	0
Single edge	3	t
Double edge	4	$\sqrt{2}\,t$
Triple edge	6	$\sqrt{3}\,t$

This is direct substitution into (1) for $i \neq j$. For B_n and C_n, the Coxeter graph is the path with $m_{i,i+1} = 3$ for $i < n-1$ and $m_{n-1,n} = 4$. Thus both types give the same matrices in this symmetric normalization. The arrow records root lengths, which this Coxeter-matrix construction does not require.

To construct a crystallographic arrangement in SageMath [15]:

```
A = hyperplane_arrangements.Coxeter("B3")
A.hyperplanes()
A.defining_polynomial()
E = hyperplane_arrangements.Coxeter("E6")
```

The constructor may use a redundant ambient space (for example for A_n or E_6); its essential rank is the Dynkin subscript.

Notation and limits. The ordinary braid group on n strands is of type A_{n-1}: $\mathrm{Art}(A_{n-1}) \cong \mathsf{B}_n$. It should not be confused with $\mathrm{Art}(B_n)$, the Artin group of Dynkin type B_n. Also, a general affine arrangement is not automatically an affine Dynkin arrangement. For instance, the triangle $\{y = 0, y = x, y = 1 - x\}$ used later has no common intersection, so it cannot be translated into a finite central reflection arrangement. Its equations remain valid input data without a Dynkin label.

B Scope and existing constructions

Let $\mathcal{A}$ be a finite arrangement of affine hyperplanes in $\mathbb{C}^\ell$, and put $M(\mathcal{A}) = \mathbb{C}^\ell \setminus \bigcup_{H \in \mathcal{A}} H$. In these appendices κ denotes the additive convolution parameter, not the gate eigenvalue λ used in the main text. Fix a one-dimensional direction Y. The minimal directional closure $\mathcal{B} = \mathcal{A}^Y$ adds the collision hyperplanes necessary to make the projection along Y a bundle of punctured affine lines. This closure and the corresponding middle convolution are part of the existing theory [20, 21]. The relationship between Long–Moody induction and Katz middle convolution, including applications to fiber-type arrangements, was developed in [22]. Earlier homological generalizations of Magnus, Burau, and Gassner representations occur in [23].

Two distinctions will be maintained throughout. First, the full convolution space is not automatically the middle-convolution space: the invariant subspace $K + L_\kappa$ must be divided out. Second, a representation on $M(\mathcal{B})$ does not automatically descend to $M(\mathcal{A})$. Neither passing to the middle quotient nor writing down all meridian matrices removes the second issue.

C The algebraic construction in arbitrary rank

The holonomy Lie algebra has the presentation

$$\mathfrak{h}(\mathcal{A}) = \mathrm{Lie}\langle t_H : H \in \mathcal{A}\rangle \Big/ \left\langle \left[t_H,\ \sum_{H' \supset X} t_{H'} \right] : X \in L_2(\mathcal{A}),\ H \supset X \right\rangle. \tag{16}$$

Here $L_2(\mathcal{A})$ consists of the nonempty intersections of codimension two. Thus an input is a finite-dimensional vector space E and endomorphisms A_H satisfying the displayed commutator relations. Extend the input to $\mathcal{B}$ by setting $A_D = 0$ for $D \in \mathcal{B} \setminus \mathcal{A}$.

Choose coordinates (x, y) with Y the y-direction. Write the transverse hyperplanes as $H_i = \{y = f_i(x)\}$, $1 \le i \le n$; all other hyperplanes are parallel to Y. The functions f_i are affine linear. A nonconstant difference $f_i - f_j$ defines a collision hyperplane in the base, whose inverse image is included in $\mathcal{B}$. Constant nonzero differences give no collision divisor. We identify a base hyperplane with its inverse image when indexing residues.

Set $W = E^n$ and use blocks indexed by i, with the coordinates of E inside each block. The known convolution formulas specialize to

$$C_{H_i} = \sum_{j=1}^{n} E_{ij} \otimes A_{H_j} + \kappa E_{ii} \otimes \mathrm{Id}_E, \tag{17}$$

$$C_D = I_n \otimes A_D + \sum_{\substack{i<j \\ \{f_i = f_j\} = D}} \big((E_{ii} - E_{ij}) \otimes A_{H_j} + (E_{jj} - E_{ji}) \otimes A_{H_i}\big). \tag{18}$$

These are the block form of [20, Proposition 3.1]. They define a representation of $\mathfrak{h}(\mathcal{B})$. The compatibility with the Lie-algebraic convolution functor is [21, Theorem 4.7]; constructing these matrices requires no integration of a differential equation.

The two invariant subspaces and the middle quotient are

$$K = \bigoplus_i e_i \otimes \ker A_{H_i}, \qquad L_\kappa = \ker\Big(\sum_i C_{H_i}\Big), \qquad \mathrm{mc}^Y_\kappa(E) = W/(K + L_\kappa). \tag{19}$$

For $\kappa \neq 0$,

$$L_\kappa = \{(v, \dots, v) : (\sum_i A_{H_i} + \kappa\,\mathrm{Id}_E)v = 0\}, \qquad K \cap L_\kappa = 0.$$

In particular, if $r = \dim E$, the output dimension is

$$nr - \sum_i \dim\ker A_{H_i} - \dim\ker(\sum_i A_{H_i} + \kappa\,\mathrm{Id}_E).$$

Constant intertwiners induce intertwiners on the quotient, giving

$$\mathrm{mc}^Y_\kappa : \mathrm{Rep}_{\mathrm{fd}}\,\mathfrak{h}(\mathcal{A}) \longrightarrow \mathrm{Rep}_{\mathrm{fd}}\,\mathfrak{h}(\mathcal{A}^Y). \tag{20}$$

This is an endofunctor on the original representation category when $\mathcal{A} = \mathcal{A}^Y$. Otherwise, the codomain in (20) must not be silently replaced by $\mathrm{Rep}_{\mathrm{fd}}\,\mathfrak{h}(\mathcal{A})$.

D Equal weights and the Burau pullback

Take $E = \mathbb{C}$, $A_{H_i} = \alpha$, $A_D = 0$ on every original parallel hyperplane, and $\kappa = \alpha \neq 0$. Then

$$C_{H_i} = \alpha e_i(\mathbf{1}^{\mathsf{T}} + e_i^{\mathsf{T}}), \qquad C_D = \alpha \sum_{\substack{i<j \\ \{f_i = f_j\} = D}} (e_i - e_j)(e_i - e_j)^{\mathsf{T}}. \tag{21}$$

The quotient in (19) is still explicitly checked: $K = 0$, and $L_\alpha = 0$ because $(n+1)\alpha \neq 0$. Hence the *additive* middle-convolution space has dimension n in this case.

Let $m = n + 1$ and define

$$F : M(\mathcal{B}) \longrightarrow \mathrm{Conf}_m(\mathbb{C}), \qquad F(x, y) = (f_1(x), \ldots, f_n(x), y). \tag{22}$$

All points on the right are distinct by construction of $\mathcal{B}$. On $V = \mathbb{C}^m/\mathbb{C}\mathbf{1}$, use quotient coordinates $w_i - w_m$. Let T_{ij} be the induced action of $(e_i - e_j)(e_i - e_j)^\mathsf{T}$ from $\mathbb{C}^m$ on V.

Proposition D.1. *Set $S = I_n + J_n$, where J_n is the all-ones matrix. In the above quotient coordinates,*

$$S(C_{H_i}/\alpha)S^{-1} = T_{i,m}, \qquad S((e_i - e_j)(e_i - e_j)^\mathsf{T})S^{-1} = T_{ij} \quad (i, j \leq n).$$

Consequently, the connection $d - \sum_H C_H \, d\log f_H$ is, after this constant change of basis, the pullback of $d - \alpha \sum_{i<j} T_{ij} \, d\log(z_i - z_j)$.

Proof. In quotient coordinates, $T_{i,m} = (\mathbf{1} + e_i)e_i^\mathsf{T}$. Since $Se_i = \mathbf{1} + e_i$ and $e_i^\mathsf{T} S = \mathbf{1}^\mathsf{T} + e_i^\mathsf{T}$, the first identity follows. Every $e_i - e_j$ has coordinate sum zero, so S acts as the identity on it and commutes with its outer product. The logarithmic forms pull back to $d\log(y - f_i)$ and $d\log(f_i - f_j)$, respectively. Grouping proportional differences gives (18). □

Put $q = \exp(2\pi i\alpha)$. For generic parameters, this is the homological reduced Burau representation restricted to P_m and pulled back by F_*. One sufficient regime for the period comparison used here is $\alpha \notin \mathbb{Z}$ and $m\alpha \notin \mathbb{Z}$. The relevant local system has common monodromy q at the m finite punctures; its twisted first homology has dimension $m - 1$. The connection/homology comparison is part of the standard framework reviewed in [24, §3], together with the convolution comparison of [20, Theorem 4.2].

Remark D.2. At resonant parameters, one must distinguish the full homological Burau module, its algebraic specialization, and the middle-convolution quotient. For example, trivial monodromy at infinity can contribute a boundary subspace to the topological middle quotient. The generic identification above is not a claim that all these spaces have the same rank at every special parameter.

E Closed formulas for based pure-braid matrices

Use the unreduced Burau convention with block $\left(\begin{smallmatrix} 1-q & q \\ 1 & 0 \end{smallmatrix}\right)$ for σ_i, and then pass to $\mathbb{C}^m/\mathbb{C}\mathbf{1}$. We act on column vectors; in products of paths the right-hand path is traversed first. Positive half twists are counterclockwise. Write

$$A_{ij} = \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i^2 \sigma_{i+1}^{-1} \cdots \sigma_{j-1}^{-1}.$$

Proposition E.1. *Let $n = m - 1$ and $1 \leq i < j \leq m$. In quotient coordinates,*

$$\beta_q(A_{ij}) = I_n + (q-1)u_{ij}v_{ij}^\mathsf{T}. \tag{23}$$

For $j \leq n$,

$$u_{ij} = qe_i + (q-1)\sum_{i<k<j} e_k - e_j, \qquad v_{ij} = e_i - e_j.$$

For $j = m$,

$$u_{i,m} = \sum_{k<i} e_k + (q+1)e_i + q\sum_{k>i} e_k, \qquad v_{i,m} = e_i.$$

In particular,

$$\det \beta_q(A_{ij}) = q^2, \qquad (\beta_q(A_{ij}) - I)(\beta_q(A_{ij}) - q^2 I) = 0.$$

Proof. For $i < n$, the square of the adjacent Burau block differs from I by $(q-1)(qe_i - e_{i+1})(e_i - e_{i+1})^{\mathsf{T}}$. Conjugate this rank-one factorization by $\beta_q(\sigma_{j-1}\cdots\sigma_{i+1})$. Moving the last nonzero entry successively to the right produces the intermediate coefficients $q-1$. At the last strand, the quotient generator has final column $(-1,\dots,-1,-q)^{\mathsf{T}}$; applying it gives the stated formula for $u_{i,m}$ and $v_{i,m}$. The case $i=n, j=m$ follows by squaring this quotient generator directly. Finally $v_{ij}^{\mathsf{T}}u_{ij} = q+1$, giving the determinant and quadratic identities. □

These are based group matrices, not individual exponentials of residues. For a general arrangement loop γ, one computes $\rho(\gamma) = \beta_q(F_*\gamma)$ from its braid word. Determining that word includes the paths from a common basepoint. At a multiple collision, the local braid is a full twist on the colliding cluster; different clusters and the connecting path must also be accounted for.

F A six-line example with fixed meridians

Start with $\mathcal{A} = \{y=0, y=x, y=1-x\}$ and project along y. Then

$$\mathcal{B} = \{y=0, y=x, y=1-x, x=0, x=1, x=\tfrac{1}{2}\}.$$

Choose basepoint $(x,y) = (1/4, 2)$, labelled configuration $(0, x, 1-x, y)$, and positive local circles. For the three transverse meridians keep $x = 1/4$ and take the y-stems below the real axis to just right of the relevant puncture. For the other meridians keep $y = 2$: the stems to $x = 0$ and $x = 1/2$ are real, ending respectively to the right and left; the stem to $x = 1$ passes above $x = 1/2$ and ends to the left of 1. Return along the reverse of each stem after the local circle.

With $d = q-1$, all six matrices are specified by $M_H = I_3 + d u_H v_H^{\mathsf{T}}$ and the following table.

H	$F_*(m_H)$	u_H^{T}	v_H^{T}
$y=0$	A_{14}	$(q+1, q, q)$	$(1,0,0)$
$y=x$	A_{24}	$(1, q+1, q)$	$(0,1,0)$
$y=1-x$	A_{34}	$(1,1,q+1)$	$(0,0,1)$
$x=0$	A_{12}	$(q,-1,0)$	$(1,-1,0)$
$x=1$	A_{13}	$(q, q-1, -1)$	$(1,0,-1)$
$x=1/2$	A_{23}	$(0, q, -1)$	$(0,1,-1)$

Explicitly, the original hyperplanes have matrices

$$M_{y=0} = \begin{pmatrix} q^2 & 0 & 0 \\ qd & 1 & 0 \\ qd & 0 & 1 \end{pmatrix}, \qquad M_{y=x} = \begin{pmatrix} 1 & d & 0 \\ 0 & q^2 & 0 \\ 0 & qd & 1 \end{pmatrix}, \qquad M_{y=1-x} = \begin{pmatrix} 1 & 0 & d \\ 0 & 1 & d \\ 0 & 0 & q^2 \end{pmatrix}.$$

The additional hyperplanes have matrices

$$M_{x=0} = \begin{pmatrix} 1+qd & -qd & 0 \\ -d & q & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad M_{x=1} = \begin{pmatrix} 1+qd & 0 & -qd \\ d^2 & 1 & -d^2 \\ -d & 0 & q \end{pmatrix},$$

$$M_{x=1/2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1+qd & -qd \\ 0 & -d & q \end{pmatrix}.$$

These extra matrices are nonidentity for $q \neq 1$, including $q = -1$. Thus this particular group representation does not descend to the original three-line complement for $q \neq 1$.

As consistency checks, setting $S = I_3 + J_3$ and denoting by $\widehat{C}_H$ the residue in (21) with $\alpha = 1$, one obtains

$$\left.\frac{dM_H}{dq}\right|_{q=1} = S\widehat{C}_H S^{-1} \quad \text{for all six hyperplanes.}$$

The standard pure-braid full-twist product also gives

$$M_{12}M_{13}M_{23}M_{14}M_{24}M_{34} = q^4 I_3.$$

The first identity is a first-order check, not an identification of M_H with the exponential of its residue at finite parameter.

G The largest descended quotient and its obstruction

The natural surjection $p : \mathfrak{h}(\mathcal{B}) \to \mathfrak{h}(\mathcal{A})$ kills precisely the Lie ideal generated by t_D, $D \in \mathcal{B} \setminus \mathcal{A}$. For a finite-dimensional $\mathfrak{h}(\mathcal{B})$-module V with action matrices B_H, define

$$N_0 = \sum_{D \in \mathcal{B}\setminus\mathcal{A}} \operatorname{im} B_D, \qquad N_{k+1} = N_k + \sum_{H \in \mathcal{B}} B_H N_k. \tag{24}$$

Proposition G.1. *The sequence in (24) stabilizes after finitely many strict dimension increases. If its stable value is N, then V/N is the largest quotient of V that factors through $\mathfrak{h}(\mathcal{A})$.*

Proof. The stable space is the smallest invariant subspace containing all $\operatorname{im} B_D$. Thus every additional generator acts as zero on V/N, and so does the ideal they generate. Conversely, the kernel of any quotient map on which those generators act trivially must contain this invariant subspace. Finite dimensionality gives termination. □

Lemma G.2. *For $\alpha \neq 0$, the operators $C_i = \alpha e_i(\mathbf{1}^{\mathsf{T}} + e_i^{\mathsf{T}})$ on $\mathbb{C}^n$ act irreducibly.*

Proof. The matrix whose ith row is $\mathbf{1}^{\mathsf{T}} + e_i^{\mathsf{T}}$ is $I_n + J_n$, which is invertible. Given a nonzero vector in an invariant subspace, some C_i therefore sends it to a nonzero multiple of e_i. For $j \neq i$, $C_j e_i = \alpha e_j$. Hence the subspace contains every coordinate vector. □

Corollary G.3. *For the equal-weight specialization of Section D, if $\mathcal{A}^Y \neq \mathcal{A}$, then its largest quotient descending to $\mathfrak{h}(\mathcal{A})$ is zero.*

Proof. An added hyperplane has at least one collision class of size at least two. Its residue is a nonzero multiple of the sum of the complete-graph Laplacians on its disjoint collision classes, so it is nonzero. The invariant subspace generated by its image is nonzero and hence, by Lemma G.2, equals the whole module. □

This corollary concerns Lie-algebra descent. For an actual group representation $\rho : G_{\mathcal{B}} \to \mathrm{GL}(V)$, the corresponding largest descended quotient is

$$V \big/ \operatorname{span}\{\rho(g)(\rho(m_D) - I)v : g \in G_{\mathcal{B}},\ D \in \mathcal{B} \setminus \mathcal{A},\ v \in V\}.$$

It kills additional meridians rather than additional residues. In particular, at resonant parameters the two criteria must not be identified without an additional argument.

H Reproducibility of the arrangement calculations

The accompanying `legacy_arrangement/` directory contains the Python/SymPy sources and regenerated reports. Those reports record checks of the block relations, middle quotients, invariant saturations, Fox derivatives, 34 pure generators for $m = 3, 4, 5, 6$, the six first derivatives, and the full-twist identity. All four arrangement verification scripts were rerun successfully for this public note package. The path-tracking check of the six braid words uses floating-point arithmetic and is not an interval-arithmetic certificate. The separate Coxeter verification files are described in the main text. Neither a finite word search nor floating-point matrix checks are used as proofs of density or of infinitude.

The reflection arrangements of types E_6, E_7, E_8 are not fiber-type, by the supersolvability classification [19, Theorem 1.2]. Thus results restricted to fiber-type arrangements do not alone cover these cases. Their Coxeter–Burau representations nevertheless exist independently of that restriction. The arrangement-group interpretation of Artin groups is also reviewed in [18, §2.1].

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